



Open Universal Arabic ASR Leaderboard

Yingzhi Wang¹, Anas Alhmoud¹, Muhammad Alqurishi¹

¹Research Center, Elm Company, KSA

ywang@elm.sa, aalhmoud@elm.sa, mualqurishi@elm.sa

Abstract

In recent years, the enhanced capabilities of ASR models and the emergence of multi-dialect datasets have increasingly pushed Arabic ASR model development toward an all-dialect-in-one direction. This trend highlights the need for benchmarking studies that evaluate model performance in a multi-dialect setting, providing the community with insights into models' generalization capabilities.

In this paper, we introduce Open Universal Arabic ASR Leaderboard, a continuous benchmark project for open-source universal Arabic ASR models across various multi-dialect datasets. We also provide a comprehensive analysis of the model's robustness, speaker adaptation, inference efficiency, and memory consumption. This work aims to offer the Arabic ASR community a reference for models' general performance and also establish a common evaluation framework for multi-dialectal Arabic ASR models.

Index Terms: Multi-dialect Arabic ASR, Arabic ASR Benchmark, ASR Leaderboard

1. Introduction

Arabic Automatic Speech Recognition (ASR) research faces unique challenges compared to other languages like English. The complexity of Arabic, with its rich morphology, dialectal variations, and lack of diacritics in written text, poses difficulties for both data collection and model development [1, 2]. Particularly regarding dialects, while Modern Standard Arabic (MSA) is commonly used in formal settings, there exist numerous regional dialects with notable differences in both pronunciation and vocabulary. Despite the significant challenge, in recent years, there has been a growing emergence of multi-dialect Arabic ASR models, driven by the release of large-scale multi-dialect datasets, enhancements in ASR model capabilities, and the growing prevalence of speech toolkits and communities.

Multi-dialect datasets play a crucial role in developing universal Arabic ASR models. Arabic dialects include but are not limited to Egyptian, Levantine, Gulf, and Maghrebi, each with distinct phonological, lexical, and syntactic variations. In this situation, gathering and labeling data require considerable time and effort. However, over the past few years, multi-dialectal datasets such as [3, 4, 5, 6, 7] have become increasingly available. For example, MGB-2 [3] consists of transcribed audio recordings sourced from Arabic broadcast TV programs, spanning over 1,200 hours of content and covering MSA, Egyptian, Gulf, Levantine, and North African dialects. The QASR dataset [6] includes over 2000 hours of speech samples including broadcast media and conversational contexts, supporting both MSA and prominent Arabic dialects. These datasets offer rich linguistic resources that capture the nuances of regional

dialects, laying the groundwork for multi-dialectal ASR model development.

Besides, End-to-end ASR models have made remarkable progress in recent years, driven by innovations in both model architectures and training techniques. Self-supervised models [8], such as Wav2Vec 2.0 [9] and HuBERT [10], enable representation learning on vast amounts of unlabeled data, followed by fine-tuning on limited labeled datasets to enhance performance in various downstream tasks. Conformer encoders [11], combining the strengths of convolutional and transformer architectures, have also been widely adopted for their ability to model both local and global dependencies in speech sequences. Additionally, OpenAI's Whisper series [12] has pushed ASR capabilities further by employing large-scale transformer-based models together with large-scale labeled datasets. These innovations support the development of multi-dialectal models from the perspective of modeling capabilities.

Speech toolkits and communities are also important in Arabic ASR evolution. For example, Hugging Face [13], NVIDIA NeMo [14] and SpeechBrain [15] offer a wide range of state-of-the-art ASR models, easy-to-use frameworks for training and inference, and a broad communication platform for the community. They grant researchers and developers the flexibility to train and advance their own Arabic ASR models.

With the growth of multi-dialect ASR models, there is an increasing demand for a thorough benchmark to assess their performance. However, existing benchmark studies are limited in scope, encompassing limited model types, limited Arabic dialects, and limited datasets, which restricts the generalizability of their results. For example, in [7], 4 models were evaluated on a single multi-dialect dataset. [16] evaluates whisper and wav2vec XLS-R [17] in n-shot setting on 5 datasets [18, 3, 19, 20, 21]. However, 2 of these 5 datasets are mono-dialectal, this limited coverage restricts the evaluation of the models' generalizability against the diversity of Arabic dialects. [22] evaluates 4 models on 7 Arabic datasets, 5 of which are mono-dialectal. In contrast to Arabic, the English ASR community benefits from a comprehensive leaderboard [23] featuring a wide range of models and multiple main-stream datasets, which has garnered substantial attention and serves as a key reference in the ASR community.

In this work, inspired by [23], we address the existing gap by introducing the largest Arabic ASR benchmark to date. Our benchmark is characterized by the following important features:

- Leverage a multi-dataset strategy with 6 different test corpus.
- Employ only multi-dialectal datasets covering all major Arabic dialects and their specific sub-dialects.
- Cover 16 state-of-the-art Arabic ASR models of various model types.

2. Experimental Settings

First, we introduce a diverse set of popular open-source Arabic ASR models and a collection of multi-dialectal Arabic datasets used in our experiments.

2.1. Models

To facilitate the collection of state-of-the-art Arabic ASR models, we divided them into four main categories.

Whisper: The OpenAI’s Whisper series of models have demonstrated strong capabilities across multiple languages including Arabic, of which the large models are scaled to 1.55 billion parameters. The training set also scales to 680,000 hours of multilingual and multitask supervision of which 739 hours are labeled Arabic speech. In our experiments we involved *whisper-large*¹, *whisper-large-v2*², *whisper-large-v3*³ and *whisper-large-v3-turbo*⁴. Whisper-large is the original largest version of the model, while the whisper-large-v2 model is trained for 2.5x more epochs with added regularization for improved performance. Whisper-large-v3 improves upon v2 by using 128 Mel frequency bins (up from 80) and training on a larger dataset. Whisper-large-v3-turbo is a pruned, fine-tuned version of Whisper-large-v3, reducing decoding layers from 32 to 4. For comparison in low-resource experimental settings, we also included *whisper-medium*⁵ and *whisper-small*⁶.

Conformer: Conformer is another highly efficient architecture designed for speech recognition tasks. NVIDIA NeMo provides performant conformer-based Connectionist Temporal Classification (CTC) [24] and RNN-Transducer (RNNT) [25] models pre-trained on different languages. In our experiment, we included Nvidia’s *conformer-ctc-large-arabic*⁷ which has around 120M parameters and is pre-trained on ASRSet 3.0 [26] with over 3600 hours of Arabic speech. Together with the Conformer-CTC model, NeMo also releases a 4-gram Arabic language model⁸. Therefore, we tested both greedy decoding and beam search with language model rescoring.

Additionally, we also included models based on Fast-Conformer [27] with CTC decoders, *nvidia-Parakeet-ctc-1.1b-concat*⁹ and *nvidia-Parakeet-ctc-1.1b-universal*¹⁰, which are specifically designed for efficient inference on long audio segments. Both models contain 1.1 billion parameters. The first model trains individual tokenizers for each language and combines them into a single tokenizer during training, while the second uses a universal tokenizer trained to handle all languages.

Self-supervised models: Self-supervised models like wav2vec 2.0 and HuBERT have also demonstrated state-of-

the-art performance in ASR by conducting down-stream fine-tuning. Due to the lack of officially released models in Arabic, we also gathered popular and performant models from the community and included them in our benchmark. We tried to cover different types of self-supervised models to maximize model diversity. *speechbrain/asr-wav2vec2-commonvoice-14-ar*¹¹ is a fine-tuned wav2vec 2.0 on Common Voice 14.0 released by SpeechBrain. *jonatasgrosman/wav2vec2-large-xlsr-53-arabic*¹² is a fine-tuned version of wav2vec 2.0 XLS-R [17] for Arabic ASR using Common Voice 6.1 and Arabic Speech Corpus [28]. *facebook/mms-1b-all*¹³ is an officially released pre-trained wav2vec 2.0 model supporting multilingual ASR for over 1000 languages[29], for the evaluation we used its Arabic ASR adaptor. In addition, *asafaya/hubert-large-arabic-transcribe*¹⁴ and *whitefox123/w2v-bert-2.0-arabic-4*¹⁵ are respectively fine-tuned wav2vec-bert 2.0 [30] and fine-tuned HuBERT that we selected from the community.

Multi-Task Foundation Models: In addition to the models mentioned above, we also considered multi-task and multi-modal foundation models. However, currently, this kind of model offers little support for Arabic ASR. In our experiment, we included Meta’s Seamless M4T [31]¹⁶, which supports Arabic ASR in three different dialects: MSA, Moroccan Arabic, and Egyptian Arabic. Since the language must be configured during inference, we conduct 3 separate inferences for each evaluation and report only the best result. Other models, such as Nvidia’s Canary [32], do not fully support Arabic at the time of our submission, but we plan to include them once they are properly deployed.

2.2. Datasets

The evaluation is conducted using a multi-dataset strategy to obtain robust scores for each model. We use only multi-dialectal datasets instead of separate datasets for each dialect. First, this is a more efficient method to cover a wider range of dialects and assess the model’s generalizability. Secondly, we consider it methodologically unsound to compare model performance on different dialects across datasets. For instance, if a model performs well on an MSA dataset but poorly on an Egyptian dataset, this does not necessarily indicate the model’s poor generalization to the Egyptian dialect, as factors like recording conditions or other external variables may also impact the model’s performance. The multi-dialectal datasets we have selected are as follows.

SADA: The Saudi Audio Dataset for Arabic (SADA) contains over 600 hours of high-quality audio recordings, covering a wide range of dialects commonly used in Saudi Arabia as well as other commonly used dialects such as MSA and Egyptian. The dataset includes various speaking styles, accents, and diverse acoustic conditions, making it highly suitable for ASR training and evaluation. We employ its test set of 10.7 hours and 10 different dialects.

¹<https://huggingface.co/openai/whisper-large>

²<https://huggingface.co/openai/whisper-large-v2>

³<https://huggingface.co/openai/whisper-large-v3>

⁴<https://huggingface.co/openai/whisper-large-v3-turbo>

⁵<https://huggingface.co/openai/whisper-medium>

⁶<https://huggingface.co/openai/whisper-small>

⁷https://catalog.ngc.nvidia.com/orgs/nvidia/teams/riva/models/speechtotext_ar_ar_conformer

⁸https://catalog.ngc.nvidia.com/orgs/nvidia/teams/riva/models/speechtotext_ar_ar_lm

⁹<https://catalog.ngc.nvidia.com/orgs/nvidia/teams/riva/models/parakeet-ctc-riva-1-1b-unified-ml-cs-concat>

¹⁰<https://catalog.ngc.nvidia.com/orgs/nvidia/teams/riva/models/parakeet-ctc-riva-1-1b-unified-ml-cs-universal>

¹¹<https://huggingface.co/speechbrain/asr-wav2vec2-commonvoice-14-ar>

¹²<https://huggingface.co/jonatasgrosman/wav2vec2-large-xlsr-53-arabic>

¹³<https://huggingface.co/facebook/mms-1b-all>

¹⁴<https://huggingface.co/asafaya/hubert-large-arabic-transcribe>

¹⁵<https://huggingface.co/whitefox123/w2v-bert-2.0-arabic-4>

¹⁶<https://huggingface.co/facebook/seamless-m4t-v2-large>

Table 1: *The Open Universal Arabic ASR Leaderboard. Apart from the WER/CER for each test set, we also report the Average WER/CER and rank the models based on the Average WER, from lowest to highest.*

model	rank	Average		SADA(Test)		CV-18.0(Test)		MASC(Test-Clean)		MASC(Test-Noisy)		MGB-2(Test)		Casablanca(Test)	
		WER↓	CER	WER	CER	WER	CER	WER	CER	WER	CER	WER	CER	WER	CER
nvidia/conformer-ctc-large-ar (lm)	1	32.91%	13.84%	44.52%	23.76%	8.80%	2.77%	23.74%	5.63%	34.29%	11.07%	17.20%	6.87%	68.9%	32.97%
nvidia/conformer-ctc-large-ar (greedy)	2	34.74%	13.37%	47.26%	22.54%	10.60%	3.05%	24.12%	5.63%	35.64%	11.02%	19.69%	7.46%	71.13%	30.5%
openai/whisper-large-v3	3	36.86%	17.21%	55.96%	34.62%	17.83%	5.74%	24.66%	7.24%	34.63%	12.89%	16.26%	7.74%	71.81%	35.04%
facebook/seamless-m4t-v2-large	4	38.16%	17.03%	62.52%	37.61%	21.70%	6.24%	25.04%	7.19%	33.24%	11.92%	20.23%	9.37%	66.25%	29.85%
openai/whisper-large-v3-turbo	5	40.05%	18.87%	60.36%	37.67%	25.73%	10.89%	25.51%	7.55%	37.16%	13.93%	17.75%	8.34%	73.79%	34.83%
openai/whisper-large-v2	6	40.20%	19.55%	57.46%	36.59%	21.77%	7.44%	27.25%	8.28%	38.55%	15.49%	25.17%	13.48%	71.01%	36.00%
openai/whisper-large	7	42.57%	20.49%	63.24%	40.16%	26.04%	9.61%	28.89%	9.05%	40.79%	16.31%	24.28%	12.10%	72.18%	35.71%
asafaya/hubert-large-ft	8	45.50%	17.35%	67.82%	31.83%	8.01%	2.37%	32.94%	7.15%	50.16%	15.62%	37.51%	11.07%	76.53%	36.03%
openai/whisper-medium	9	45.57%	22.27%	67.71%	43.83%	28.07%	10.38%	29.99%	8.98%	42.91%	17.49%	29.32%	14.82%	75.44%	38.12%
nvidia-Parakeet-ctc-1.1b-concat	10	46.54%	23.88%	70.70%	46.70%	26.34%	9.82%	30.49%	8.41%	45.95%	18.72%	24.94%	9.87%	80.80%	49.77%
nvidia-Parakeet-ctc-1.1b-universal	11	51.96%	25.19%	73.58%	49.48%	40.01%	14.64%	36.16%	10.29%	50.03%	20.09%	30.68%	11.36%	81.30%	45.31%
facebook/mms-1b-all	12	54.54%	21.45%	77.48%	37.50%	26.52%	7.21%	38.82%	10.36%	57.33%	19.76%	39.16%	13.48%	87.95%	40.41%
whitefox123/w2v-bert-2.0-ft	13	55.13%	21.68%	78.02%	33.17%	24.18%	6.79%	35.93%	9.01%	56.36%	19.43%	48.64%	15.56%	87.64%	46.12%
openai/whisper-small	14	58.13%	27.62%	87.34%	56.75%	41.79%	15.75%	37.82%	11.92%	53.28%	21.93%	40.66%	19.39%	87.88%	39.99%
jonatasgrosman/wav2vec2-large-xlsr-ft	15	60.98%	25.61%	86.82%	44.20%	23.00%	6.64%	42.75%	11.87%	64.27%	24.17%	56.29%	20.44%	92.72%	46.33%
speechbrain/wav2vec2-large-ft	16	65.74%	30.93%	88.54%	50.28%	29.17%	9.85%	49.10%	16.37%	69.57%	30.17%	64.37%	26.56%	93.68%	52.36%

Common Voice: The Common Voice 18.0 version contains an Arabic subset with 91 hours of validated Arabic speech, including many common dialects such as Egyptian, Syrian, North African, etc. We use its test set of 12.6 hours and 25 different dialects. It should be noted that Common Voice shuffles the dataset splits from one release to another, the results may be biased for those models trained on previous versions due to the test data leakage, but according to the results, this will not significantly impact the overall benchmark.

MASC: The Massive Arabic Speech Corpus (MASC) is a comprehensive dataset that contains 1,000 hours of speech recordings from over 700 YouTube channels. The dataset is multi-regional, multi-genre, and multi-dialect. Besides MSA, it covers more than 20 dialects that are commonly used in Syria, Egypt, Saudi Arabia, Jordan, etc. We evaluate the models with the test-clean and test-noisy subsets, the previous has 10.5 hours and 7 dialects, while the latter has 8.9 hours and 14 dialects.

MGB-2: The MGB-2 dataset is a large-scale speech corpus developed for the Arabic language, it contains approximately 1,200 hours of broadcast TV data, spanning a wide range of genres such as news, talk shows, and drama, recorded from 5 different Arabic dialects. In our benchmark, we adopted its 9.6-hour test set.

Casablanca: Casablanca is a large-scale community-driven multi-dialectal Arabic dataset. The dataset covers eight dialects: Algerian, Egyptian, Emirati, Jordanian, Mauritanian, Moroccan, Palestinian, and Yemeni, and includes annotations for transcription, gender, dialect, and code-switching. In our benchmark, we utilized its 8-hour test set.

2.3. Metrics

We compute and report both Word Error Rate (WER) and Character Error Rate (CER) in the leaderboard. For text normalization, we follow the work in [33], removing all punctuation as well as the diacritics, and normalizing characters with Hamzas and Maddas. We also convert all Eastern Arabic numerals to Western Arabic numerals. Since certain models might support code-switching and the test sets contain very few code-switching samples, we kept the Latin characters and included code-switching errors in the overall error computation.

3. Results and Analysis

3.1. Overall Leaderboard

The overall leaderboard is shown in Table 1. For all the collected models, we ran inference across six different multi-

dialect test sets. We then averaged the WER and CER over the test sets and ranked the models based on the average WER. As can be seen from the results, Nvidia’s Conformer-CTC model, combined with its released 4-gram language model, ranks first on the leaderboard with 25.71% Average WER and 10.02% Average CER. The Whisper large series and seamless-m4t model follow closely behind, while smaller Whisper models and fine-tuned self-supervised models rank last. Moreover, we find that the model rankings are closely aligned with the scale of the training datasets. For example, the top-ranked Nvidia Conformer was trained on 3,600 hours of labeled data, the middle-ranked whisper models were trained on more than 700 hours of labeled data, whereas the lower-ranked self-supervised models were only fine-tuned with less than 100 hours of labeled data, making them difficult to generalize well to all the test sets.

Table 2: *Model performance (WER) under 3 different acoustic conditions using SADA dataset. The models are sorted based on the WER under the clean environment from lowest to highest.*

model	Clean↓	Noisy	Music
nvidia/conformer-ctc-large-ar (lm)	37.83%	49.21%	47.30%
nvidia/conformer-ctc-large-ar (greedy)	40.52%	51.64%	50.51%
openai/whisper-large-v2	51.21%	60.54%	61.76%
openai/whisper-large-v3	51.45%	58.47%	58.71%
openai/whisper-large-v3-turbo	52.96%	63.51%	66.08%
openai/whisper-large	55.73%	70.26%	64.04%
facebook/seamless-m4t-v2-large	56.98%	66.48%	64.69%
openai/whisper-medium	60.82%	73.23%	69.68%
asafaya/hubert-large-ft	62.80%	70.72%	70.68%
nvidia-Parakeet-ctc-1.1b-concat	64.01%	75.94%	72.73%
nvidia-Parakeet-ctc-1.1b-universal	66.10%	78.59%	76.93%
facebook/mms-1b-all	72.04%	80.92%	80.15%
whitefox123/w2v-bert-2.0-ft	74.92%	79.88%	79.68%
openai/whisper-small	77.48%	91.17%	95.47%
jonatasgrosman/wav2vec2-large-xlsr-ft	82.62%	89.14%	89.35%
speechbrain/wav2vec2-large-ft	85.21%	90.53%	90.34%

3.2. Robustness

Furthermore, we assessed the model’s robustness under different acoustic conditions and across different dialects within the SADA dataset, the results are presented in Table 2 and Table 3. Under noisy or music environments, all models show a certain degree of performance degradation compared to the clean environment. Similarly, Nvidia’s Conformer-CTC-Large model combined with its language model still tops the list with the best WER under all environmental conditions, followed by the

Whisper large series and self-supervised models.

As for the dialect variation, the results suggest that all models achieve their best results on MSA, but exhibit a significant decline when applied to dialects such as Egyptian and Khaliji. This effect is likely caused by the imbalanced distribution of dialects in most public datasets, with MSA generally representing a higher percentage than other dialects.

Table 3: *Model performance (WER) across 5 different dialects using SADA dataset. The models are sorted by the WER on MSA from lowest to highest.*

model	MSA↓	Egyptian	Hijazi	Khaliji	Najdi
nvidia/conformer-ctc-large-ar (lm)	19.23%	40.97%	36.96%	48.23%	36.34%
nvidia/conformer-ctc-large-ar (greedy)	21.56%	41.97%	39.84%	51.12%	38.70%
openai/whisper-large-v3	27.95%	59.28%	49.99%	59.92%	48.58%
openai/whisper-large-v3-turbo	28.93%	63.01%	54.35%	65.04%	52.42%
openai/whisper-large-v2	29.13%	48.44%	51.40%	63.06%	51.49%
openai/whisper-large	31.89%	69.12%	59.67%	64.71%	58.01%
nvidia-Parakeet-ctc-1.1b-concat	36.93%	66.00%	62.93%	76.68%	60.91%
nvidia-Parakeet-ctc-1.1b-universal	37.34%	63.89%	66.62%	79.39%	62.95%
openai/whisper-medium	38.31%	55.79%	60.83%	71.30%	63.97%
asafaya/hubert-large-ft	39.51%	74.47%	63.05%	73.12%	61.07%
facebook/seamless-m4t-v2-large	39.87%	72.98%	54.02%	55.73%	54.89%
facebook/mms-1b-all	42.71%	81.94%	73.21%	83.43%	69.95%
openai/whisper-small	49.41%	69.61%	86.62%	93.16%	87.69%
whitefox123/w2v-bert-2.0-ft	51.13%	90.29%	73.60%	80.08%	75.62%
jonatasgrosman/wav2vec2-large-xlsr-ft	61.51%	89.79%	84.48%	90.32%	82.16%
speechbrain/wav2vec2-large-ft	66.06%	91.53%	86.50%	91.10%	85.26%

Table 4: *Model performance (WER) across Speakers' age and gender; the models are sorted based on the WER for adult speakers from lowest to highest.*

model	Age		Gender	
	Adult↓	Elderly	Female	Male
nvidia/conformer-ctc-large-ar (lm)	37.85%	44.31%	40.19%	37.91%
nvidia/conformer-ctc-large-ar (greedy)	40.51%	44.41%	43.04%	40.46%
openai/whisper-large-v3	50.01%	55.28%	54.32%	49.60%
openai/whisper-large-v2	52.25%	52.51%	55.48%	52.03%
openai/whisper-large-v3-turbo	53.74%	57.23%	55.79%	54.07%
facebook/seamless-m4t-v2-large	54.46%	51.38%	48.84%	55.53%
openai/whisper-large	58.06%	58.67%	62.40%	57.70%
asafaya/hubert-large-ft	62.45%	70.36%	70.03%	61.64%
openai/whisper-medium	62.47%	60.72%	68.62%	61.66%
nvidia-Parakeet-ctc-1.1b-concat	63.01%	71.90%	67.90%	62.64%
nvidia-Parakeet-ctc-1.1b-universal	65.28%	75.08%	71.91%	64.65%
facebook/mms-1b-all	71.28%	80.92%	80.87%	70.12%
whitefox123/w2v-bert-2.0-ft	73.99%	78.15%	77.26%	73.70%
jonatasgrosman/wav2vec2-large-xlsr-ft	82.66%	88.72%	88.68%	81.95%
speechbrain/wav2vec2-large-ft	85.08%	89.95%	89.31%	84.63%
openai/whisper-small	85.29%	92.10%	94.05%	84.59%

3.3. Speaker Adaptation

In order to ensure benchmark fairness and showcase evaluation bias, we also conducted an analysis of speaker adaptation on the SADA dataset, taking both the speaker's age and gender into account. The results indicate that for most of the models, elderly speakers generally have higher WER compared to adults, and female speakers tend to have slightly higher WER than male speakers, suggesting a common bias of age and gender in Arabic ASR models.

3.4. Efficiency and Resource Usage

Given the importance of efficiency for production ASR models, we also evaluated the efficiency of each model on the SADA dataset. This includes the real-time factor (RTF) on GPU and the GPU memory usage during inference, covering both memory for model loading and average inference memory per sample. Instead of running tests on a full-load setup as in [23], we

tried to avoid repeated work and used a minimal-load configuration. Namely, we used a batch size of 1 with an average audio duration of 6.25 seconds, which aligns more closely with real-world industrial scenarios. The RTFs are measured on a single A100 40GB GPU with 5 warm-up runs.

The results reveal that, in conditions of small batch size and short audio duration, self-supervised models achieve higher efficiency, exhibiting lower RTF and reduced memory usage. In contrast, large Whisper models are significantly slower and consume more memory. The Conformer-CTC model, meanwhile, demonstrates impressive speed and requires minimal memory.

Table 5: *Models' efficiency and resource usage on the SADA dataset with minimal batch load. The models are sorted by RTF from lowest to highest. #Mem_model refers to the memory usage (GB) with only the model loaded, while #Mem_infer refers to the average inference-only memory (GB) per sample.*

model	RTF↓	#Mem_model	#Mem_infer
jonatasgrosman/wav2vec2-large-xlsr-ft	0.0064	1.21	0.14
facebook/mms-1b-all	0.0137	3.69	0.14
speechbrain/wav2vec2-large-ft	0.0137	1.23	0.14
asafaya/hubert-large-ft	0.0142	1.23	0.14
nvidia/conformer-ctc-large-ar (greedy)	0.0470	0.48	0.07
nvidia/conformer-ctc-large-ar (lm)	0.0764	0.48	0.07
whitefox123/w2v-bert-2.0-ft	0.1168	2.26	0.14
openai/whisper-small	0.1791	0.48	0.12
openai/whisper-medium	0.2314	1.43	0.15
facebook/seamless-m4t-v2-large	0.2491	5.62	0.20
openai/whisper-large-v3-turbo	0.3066	1.52	0.19
openai/whisper-large-v3	0.4335	2.88	0.24
openai/whisper-large	0.4758	2.88	0.24
openai/whisper-large-v2	0.4789	2.88	0.24

3.5. Discussions

As a large-scale, comprehensive benchmark, this study has several limitations that are open to discussion. First, similar to [23], we tried to minimize the in-domain issue based on known information. However, given the limited language resources, we included the Common Voice dataset and some models fine-tuned on it. As our benchmark includes multiple datasets, this effect on the overall rankings is reduced. Additionally, the dataset inherently suffers from unavoidable attribute distribution imbalances, for example in the MGB-2, over 70% of the samples represent the MSA dialect, which can potentially bias the results.

This benchmark is designed as a continuous and long-term project. We plan to regularly update the models and datasets alongside Arabic ASR progress to provide up-to-date references for the community. The leaderboard is hosted on Hugging-Face¹⁷ and the evaluation code is open-sourced on GitHub¹⁸.

4. Conclusions

In this work, we implement the most comprehensive Arabic ASR benchmark to date, which includes an Open Universal Arabic ASR Leaderboard and a thorough analysis covering robustness, speaker adaptation, efficiency, and resource utilization. As a continuous effort, our work aims to provide the Arabic ASR community with the latest references on model performance and develop a standard evaluation framework for Arabic ASR models.

¹⁷https://huggingface.co/spaces/elmresearchcenter/open_universal_arabic_asr_leaderboard

¹⁸https://github.com/Natural-Language-Processing-Elm/open_universal_arabic_asr_leaderboard

5. References

- [1] A. Dhoub, A. Othman, O. El Ghoul, M. K. Khribi, and A. Al Sinani, "Arabic automatic speech recognition: a systematic literature review," *Applied Sciences*, vol. 12, no. 17, p. 8898, 2022.
- [2] A. A. Abdelhamid, H. A. Alsayadi, I. Hegazy, and Z. T. Fayed, "End-to-end arabic speech recognition: A review," in *Proceedings of the 19th Conference of Language Engineering (ESOLEC'19)*, Alexandria, Egypt, 2020, pp. 26–30.
- [3] A. Ali, P. Bell, J. Glass, Y. Messaoui, H. Mubarak, S. Renals, and Y. Zhang, "The mgb-2 challenge: Arabic multi-dialect broadcast media recognition," in *2016 IEEE Spoken Language Technology Workshop (SLT)*. IEEE, 2016, pp. 279–284.
- [4] S. Alharbi, A. Alowisheq, Z. Tüske, K. Darwish, A. Alrajeh, A. Alrowithi, A. B. Tamran, A. Ibrahim, R. Aloraini, R. Alnajim *et al.*, "Sada: Saudi audio dataset for arabic," in *ICASSP 2024-2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2024, pp. 10 286–10 290.
- [5] M. Al-Fetyani, M. Al-Barham, G. Abandah, A. Alsharkawi, and M. Dawas, "Masc: Massive arabic speech corpus," in *2022 IEEE Spoken Language Technology Workshop (SLT)*. IEEE, 2023, pp. 1006–1013.
- [6] H. Mubarak, A. Hussein, S. A. Chowdhury, and A. Ali, "Qasr: Qeri aljazeera speech resource—a large scale annotated arabic speech corpus," *arXiv preprint arXiv:2106.13000*, 2021.
- [7] B. Talafha, K. Kadaoui, S. M. Magdy, M. Habiboullah, C. M. Chafei, A. O. El-Shangiti, H. Zayed, R. Alhamouri, R. Assi, A. Alraeesi *et al.*, "Casablanca: Data and models for multidialectal arabic speech recognition," *arXiv preprint arXiv:2410.04527*, 2024.
- [8] A. Baevski, W.-N. Hsu, Q. Xu, A. Babu, J. Gu, and M. Auli, "Data2vec: A general framework for self-supervised learning in speech, vision and language," in *International Conference on Machine Learning*. PMLR, 2022, pp. 1298–1312.
- [9] A. Baevski, Y. Zhou, A. Mohamed, and M. Auli, "wav2vec 2.0: A framework for self-supervised learning of speech representations," *Advances in neural information processing systems*, vol. 33, pp. 12 449–12 460, 2020.
- [10] W.-N. Hsu, B. Bolte, Y.-H. H. Tsai, K. Lakhotia, R. Salakhutdinov, and A. Mohamed, "Hubert: Self-supervised speech representation learning by masked prediction of hidden units," *IEEE/ACM transactions on audio, speech, and language processing*, vol. 29, pp. 3451–3460, 2021.
- [11] A. Gulati, J. Qin, C.-C. Chiu, N. Parmar, Y. Zhang, J. Yu, W. Han, S. Wang, Z. Zhang, Y. Wu *et al.*, "Conformer: Convolution-augmented transformer for speech recognition," *arXiv preprint arXiv:2005.08100*, 2020.
- [12] A. Radford, J. W. Kim, T. Xu, G. Brockman, C. McLeavey, and I. Sutskever, "Robust speech recognition via large-scale weak supervision," in *International conference on machine learning*. PMLR, 2023, pp. 28 492–28 518.
- [13] T. Wolf, "Huggingface's transformers: State-of-the-art natural language processing," *arXiv preprint arXiv:1910.03771*, 2019.
- [14] O. Kuchaiev, J. Li, H. Nguyen, O. Hrinchuk, R. Leary, B. Ginsburg, S. Krizan, S. Beliaev, V. Lavrukhin, J. Cook *et al.*, "Nemo: a toolkit for building ai applications using neural modules," *arXiv preprint arXiv:1909.09577*, 2019.
- [15] M. Ravanelli, T. Parcollet, A. Moumen, S. de Langen, C. Subakan, P. Plantinga, Y. Wang, P. Mousavi, L. Della Libera, A. Ploujnikov *et al.*, "Open-source conversational ai with speechbrain 1.0," *Journal of Machine Learning Research*, vol. 25, no. 333, pp. 1–11, 2024.
- [16] B. Talafha, A. Waheed, and M. Abdul-Mageed, "N-shot benchmarking of whisper on diverse arabic speech recognition," *arXiv preprint arXiv:2306.02902*, 2023.
- [17] A. Conneau, A. Baevski, R. Collobert, A. Mohamed, and M. Auli, "Unsupervised cross-lingual representation learning for speech recognition," *arXiv preprint arXiv:2006.13979*, 2020.
- [18] R. Ardila, M. Branson, K. Davis, M. Henretty, M. Kohler, J. Meyer, R. Morais, L. Saunders, F. M. Tyers, and G. Weber, "Common voice: A massively-multilingual speech corpus," *arXiv preprint arXiv:1912.06670*, 2019.
- [19] A. Ali, S. Vogel, and S. Renals, "Speech recognition challenge in the wild: Arabic mgb-3," in *2017 IEEE Automatic Speech Recognition and Understanding Workshop (ASRU)*. IEEE, 2017, pp. 316–322.
- [20] A. Ali, S. Shon, Y. Samih, H. Mubarak, A. Abdelali, J. Glass, S. Renals, and K. Choukri, "The mgb-5 challenge: Recognition and dialect identification of dialectal arabic speech," in *2019 IEEE Automatic Speech Recognition and Understanding Workshop (ASRU)*. IEEE, 2019, pp. 1026–1033.
- [21] A. Conneau, M. Ma, S. Khanuja, Y. Zhang, V. Axelrod, S. Dalmia, J. Riesa, C. Rivera, and A. Bapna, "Fleurs: Few-shot learning evaluation of universal representations of speech," in *2022 IEEE Spoken Language Technology Workshop (SLT)*. IEEE, 2023, pp. 798–805.
- [22] A. Abdelali, H. Mubarak, S. Chowdhury, M. Hasanain, B. Mousi, S. Boughorbel, S. Abdaljalil, Y. El Kheir, D. Izham, F. Dalvi *et al.*, "Larabench: Benchmarking arabic ai with large language models," in *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2024, pp. 487–520.
- [23] V. Srivastav, S. Majumdar, N. Koluguri, A. Moumen, S. Gandhi *et al.*, "Open automatic speech recognition leaderboard," <https://huggingface.co/spaces/hf-audio/open-asr-leaderboard>, 2023.
- [24] A. Graves, S. Fernández, F. Gomez, and J. Schmidhuber, "Connectionist temporal classification: labelling unsegmented sequence data with recurrent neural networks," in *Proceedings of the 23rd international conference on Machine learning*, 2006, pp. 369–376.
- [25] A. Graves, "Sequence transduction with recurrent neural networks," *arXiv preprint arXiv:1211.3711*, 2012.
- [26] V. Noroozi, S. Majumdar, A. Kumar, J. Balam, and B. Ginsburg, "Stateful conformer with cache-based inference for streaming automatic speech recognition," in *ICASSP 2024-2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2024, pp. 12 041–12 045.
- [27] D. Rekesh, N. R. Koluguri, S. Krizan, S. Majumdar, V. Noroozi, H. Huang, O. Hrinchuk, K. Puvvada, A. Kumar, J. Balam, and B. Ginsburg, "Fast conformer with linearly scalable attention for efficient speech recognition," in *2023 IEEE Automatic Speech Recognition and Understanding Workshop (ASRU)*, 2023, pp. 1–8.
- [28] N. Halabi *et al.*, "Arabic speech corpus," *Oxford Text Archive Core Collection*, 2016.
- [29] V. Pratap, A. Tjandra, B. Shi, P. Tomasello, A. Babu, S. Kundu, A. Elkahky, Z. Ni, A. Vyas, M. Fazel-Zarandi *et al.*, "Scaling speech technology to 1,000+ languages," *Journal of Machine Learning Research*, vol. 25, no. 97, pp. 1–52, 2024.
- [30] Y.-A. Chung, Y. Zhang, W. Han, C.-C. Chiu, J. Qin, R. Pang, and Y. Wu, "W2v-bert: Combining contrastive learning and masked language modeling for self-supervised speech pre-training," in *2021 IEEE Automatic Speech Recognition and Understanding Workshop (ASRU)*. IEEE, 2021, pp. 244–250.
- [31] L. Barrault, Y.-A. Chung, M. C. Meglioli, D. Dale, N. Dong, P.-A. Duquenne, H. Elshahar, H. Gong, K. Heffernan, J. Hoffman *et al.*, "Seamlessm4t-massively multilingual & multimodal machine translation," *arXiv preprint arXiv:2308.11596*, 2023.
- [32] K. C. Puvvada, P. Żelasko, H. Huang, O. Hrinchuk, N. R. Koluguri, K. Dhawan, S. Majumdar, E. Rastorgueva, Z. Chen, V. Lavrukhin *et al.*, "Less is more: Accurate speech recognition & translation without web-scale data," *arXiv preprint arXiv:2406.19674*, 2024.
- [33] S. A. Chowdhury, A. Hussein, A. Abdelali, and A. Ali, "Towards one model to rule all: Multilingual strategy for dialectal code-switching arabic asr," *arXiv preprint arXiv:2105.14779*, 2021.